

FIRST ANNUAL PROGRESS SEMINAR REPORT

ON

LIE'S THIRD THEOREM

Presented by:
DIBENDU PAL

Supervised by:
PROF. SHRIPAD GARGE

Roll - 23D0785

September 26 , 2024



Department of Mathematics
Indian Institute of Technology, Bombay
Powai, Mumbai-400076

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1 Basic Prerequisites

In this section we discuss some basic theorems and lemmas we need for later section.

Definition. For an $n \times n$ matrix A , define $\log A$ by

$$\log A = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{(A - I)^m}{m}$$

if the series converges. Since the complex-valued series of the above series has radius of convergence 1 and since $\|(A - I)^m\| \leq \|A - I\|^m$ for $m \geq 1$, the matrix-valued series will converge absolutely if $\|A - I\| < 1$. Even if $\|A - I\| > 1$, the series converges if $A - I$ is nilpotent

Theorem 1.1. *The function*

$$\log A = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{(A - I)^m}{m}$$

is defined and continuous on the set of all $n \times n$ complex matrices A with $\|A - I\| < 1$. For all A with $\|A - I\| < 1$,

$$e^{\log A} = A$$

For all X with $\|X\| < \log 2$, $\|e^X - I\| < 1$ and

$$\log e^X = X$$

Proof. Since the series above converges absolutely and uniformly for all A with $\|A - I\| < 1$, we get that the function is continuous w.r.t A in the interval.

Now define $\log z$ with the same above series which defines a holomorphic function on $|z - 1| < 1$ and coincides with usual logarithm for real z in $(0, 2)$. Now, $\exp(\log z) = z$ for $z \in (0, 2)$ and since both sides of this are holomorphic in z , the identity theorem allows them to continue to hold on the whole set $\{|z - 1| < 1\}$. On the other hand, if $|z| < \log 2$, then

$$|e^z - 1| = \left| z + \frac{z^2}{2!} + \dots \right| \leq |z| + \frac{|z|^2}{2!} + \dots = e^{|z|} - 1 < 1$$

Thus, $\log(\exp z)$ is defined for all such z . Since $\log(\exp z) = u$ for real z with $|z| < \log 2$, it follows that $\log(\exp u) = u$ for all complex numbers with $|z| < \log 2$ by the identity theorem.

Assume now that A satisfies $\|A - I\| < 1$. If A is diagonalizable with eigenvalue $z_1, \dots, z_n$, then we can express A in the form CDC^{-1} with D diagonal, in which case

$$(A - I)^m = C \begin{pmatrix} (z_1 - 1)^m & & 0 \\ & \ddots & \\ 0 & & (z_n - 1)^m \end{pmatrix} C^{-1}$$

Since $|\lambda| < \|A\|$, for eigenvalue λ for matrix A , we get $|z_j - 1| < \|A - I\| < 1$. Thus, the Matrix Logarithm

$$e^{\log A} = C \begin{pmatrix} e^{\log z_1} & & 0 \\ & \ddots & \\ 0 & & e^{\log z_n} \end{pmatrix} C^{-1} = A$$

If A is not diagonalizable, we approximate A by a sequence A_m of diagonalizable matrices and use the continuity of the logarithm and exponential functions. Thus, $\exp(\log A) = A$ for all A with $\|A - I\| < 1$.

The same argument as in the complex case shows that if $\|X\| < \log 2$, then $\|e^X - I\| < 1$. The proof that $\log(e^X) = X$ is then similar to the proof that $\exp(\log A) = A$. $\square$

Definition. Let G be a matrix lie group. The lie algebra of G denoted as $\mathfrak{g}$ is the set of all matrices X s.t $e^{tX} \in G$ for all real numbers t .

Lemma 1.2. Assume $G \subset \text{GL}(m; \mathbb{C})$ is a matrix Lie group with Lie algebra $\mathfrak{g}$. Assume B_n is a sequence in G and $B_n \rightarrow I$. Let $Y_n = \log B_n$, which is defined for all large n . Suppose that Y_n is nonzero for all n and $Y_n / \|Y_n\| \rightarrow Y \in M_n(\mathbb{C})$. Then $Y \in \mathfrak{g}$.

Proof. For any $t \in \mathbb{R}$, we have $(t / \|Y_n\|) Y_n \rightarrow tY$. Observe that since $B_n \rightarrow I$, we get $\|Y_n\| \rightarrow 0$. Thus, we can find integers k_n such that $k_n \|Y_n\| \rightarrow t$. Denote $\|Y_n\|$ as a_n . Then, take $k_n = \lfloor \frac{t}{a_n} \rfloor$. Using squeeze theorem for the inequality,

$$a_n \left(\frac{t}{a_n} - 1 \right) \leq a_n \left\lfloor \frac{t}{a_n} \right\rfloor \leq a_n \left(\frac{t}{a_n} \right)$$

we get that $k_n a_n \rightarrow t$. We get,

$$e^{k_n Y_n} = \exp \left[(k_n \|Y_n\|) \frac{Y_n}{\|Y_n\|} \right] \rightarrow e^{tY}$$

But,

$$e^{k_n Y_n} = (e^{Y_n})^{k_n} = (B_n)^{k_n} \in G$$

and G is closed. So, $e^{tY} \in G$. Hence, $Y \in \mathfrak{g}$. $\square$

Theorem 1.3. For $\varepsilon \in (0, \log 2)$, let $U_\varepsilon = \{X \in M_n(\mathbb{C}) \mid \|X\| < \varepsilon\}$ and let $V_\varepsilon = \exp(U_\varepsilon)$. Suppose $G \subset \text{GL}(n; \mathbb{C})$ is a matrix Lie group with Lie algebra $\mathfrak{g}$. Then there exists $\varepsilon \in (0, \log 2)$ such that for all $A \in V_\varepsilon$, A is in G if and only if $\log A$ is in $\mathfrak{g}$.

Proof. Observe that for all $A \in V_\varepsilon$, $A = e^X$ given $\|X\| < \log 2$, which implies $\|A - I\| < 1$. So, $\log(e^X) = X$. Now if $X = \log A$ is in $\mathfrak{g}$, then $A = e^{\log A} = e^X$ is in G . So, need to prove the other direction.

Let us view $M_n(\mathbb{C})$ as $\mathbb{C}^{n^2} \cong \mathbb{R}^{2n^2}$ and say D be the orthogonal complement of $\mathfrak{g}$

with respect to the normal inner product on $\mathbb{R}^{2n^2}$. Consider the map $\Phi : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$ given by

$$\Phi(Z) = e^X e^Y$$

where $Z = X + Y$ with $X \in \mathfrak{g}$ and $Y \in D$. We claim that derivative of ϕ at origin $D_0(\Phi) = I$. Take $h = h_1 + h_2$ where $h_1 \in X, h_2 \in Y$. Then,

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\|e^{h_1} \cdot e^{h_2} - I - Ih\|}{\|h\|} &= \lim_{h \rightarrow 0} \|h_1 h_2 + \frac{h_1^2}{2!} + \frac{h_2^2}{2!} + \dots\| \frac{1}{\|h\|} \\ &\leq \lim_{h \rightarrow 0} \left(\|h\|^2 + \frac{\|h\|^2}{2!} + \frac{\|h\|^2}{2!} + \dots \right) \frac{1}{\|h\|} \\ &= \lim_{h \rightarrow 0} \left(\|h\| + \frac{\|h\|}{2!} + \frac{\|h\|}{2!} + \dots \right) = 0 \end{aligned}$$

Since the derivative of Φ at the origin is invertible, applying inverse function theorem we get Φ has a continuous local inverse, defined in a neighborhood of I . We need to prove that for some ε , if $A \in V_\varepsilon \cap G$, then $\log A \in \mathfrak{g}$. If this is not true, we can find a sequence A_m in G such that $A_m \rightarrow I$ as $m \rightarrow \infty$ and such that for all $m, \log A_m \notin \mathfrak{g}$. Using the local inverse of the map Φ , we can write A_m (for all sufficiently large m) as

$$A_m = e^{X_m} e^{Y_m}, \quad X_m \in \mathfrak{g}, Y_m \in D$$

with $X_m, Y_m \rightarrow 0$ as $m \rightarrow \infty$. We must have $Y_m \neq 0$, otherwise we would get $\log A_m = X_m \in \mathfrak{g}$. Since $e^{X_m}, A_m \in G$, we see that

$$B_m := e^{-X_m} A_m = e^{Y_m} \in G.$$

Since the unit sphere in D is compact, we can choose a subsequence of the Y_m so that $Y_m / \|Y_m\|$ converges to some $Y \in D$, with $\|Y\| = 1$. By the lemma (1.2), $Y \in \mathfrak{g}$. This is a contradiction, as D is the orthogonal complement of $\mathfrak{g}$. So, there must be some ε such that $\log A \in \mathfrak{g}$ for all A in $V_\varepsilon \cap G$. $\square$

Theorem 1.4. *Let $X(t)$ be a smooth matrix valued function. Then*

$$\frac{d}{dt} e^{X(t)} = e^{X(t)} \left\{ \frac{I - e^{-ad_{X(t)}}}{ad_{X(t)}} \left(\frac{dX}{dt} \right) \right\}$$

Proof. Write X instead of $X(t)$, and let $Y = X'$ be the derivative. Then using the product rule of derivative we have :

$$\begin{aligned} I' &= (X^0)' = 0, \\ (X^1)' &= Y, \\ (X^2)' &= YX + XY, \\ (X^3)' &= YXX + XYX + XXY, \\ (X^4)' &= YXXX + XXYX + XYXX + XXXY, \\ (X^{n+1})' &= YX^n + XYX^{n-1} + \dots + X^n Y = \sum_{\substack{j,k \geq 0 \\ j+k=n}} X^j Y X^k. \end{aligned}$$

So,

$$\begin{aligned}
\frac{d}{dt}e^{X(t)} &= Y + \frac{1}{2!}(YX + XY) + \frac{1}{3!}(YX^2 + XYX + X^2Y) + \dots \\
&= \sum_{k=0}^{\infty} \sum_{r=0}^k \frac{1}{(k+1)!} X^r Y X^{k-r} \\
&= \sum_{k=0}^{\infty} \sum_{r=0}^k \frac{B(k-r+1, r+1)}{r!(k-r)!} X^r Y X^{k-r} \\
&= \sum_{k=0}^{\infty} \sum_{r=0}^k \frac{\int_0^1 s^r (1-s)^{k-r} ds}{r!(k-r)!} X^{k-r} Y X^r \\
&= \int_0^1 \sum_{k=0}^{\infty} \sum_{r=0}^k \frac{(1-s)^{k-r} s^r}{r!(k-r)!} X^{k-r} Y X^r ds \quad (D.C.T) \\
&= \int_0^1 \left(\sum_{n=0}^{\infty} \frac{1}{n!} (1-s)^n X^n \right) Y \left(\sum_{n=0}^{\infty} \frac{1}{n!} s^n X^n \right) ds \\
&= \int_0^1 e^{(1-s)X} Y e^{sX} ds.
\end{aligned}$$

Thus,

$$\begin{aligned}
\int_0^1 e^{(1-s)X} Y e^{sX} ds &= e^X \int_0^1 Ad_{e^{-sX}}(Y) ds \\
&= e^X \int_0^1 e^{-ad_{sX}}(Y) ds \\
&= e^X \left(\int_0^1 e^{-ad_{sX}} ds \right) (Y) \\
&= e^X \left(\int_0^1 \sum_{k=0}^{\infty} \frac{(-1)^k s^k}{k!} (ad_X)^k ds \right) (Y) \\
&= e^X \left\{ \frac{1 - e^{-ad_X}}{ad_X} (Y) \right\}
\end{aligned}$$

□

2 The Baker-Campbell-Haursdorff Formula

The goal of this section is to compute $\log(e^X e^Y)$ in terms of their commutators when X, Y are small. Consider the function

$$g(z) = \frac{\log z}{1 - \frac{1}{z}}$$

which is defined and holomorphic in the disk $\{|z - 1| < 1\}$. Thus, $g(z)$ can be expressed as a series

$$g(z) = \sum_{m=0}^{\infty} a_m (z - 1)^m = g(z) = 1 + \frac{1}{2}(z - 1) - \frac{1}{6}(z - 1)^2 + \frac{1}{12}(z - 1)^3 - \dots$$

with radius of convergence one. If V is a finite-dimensional vector space over $\mathbb{C}$ then for any operator A on V with $\|A - I\| < 1$, we can define

$$g(A) = \sum_{m=0}^{\infty} a_m (A - I)^m$$

Theorem 2.1. (Baker-Campbell-Hausdorff formula). *For all $n \times n$ complex matrices X and Y with $\|X\|$ and $\|Y\|$ sufficiently small, we have*

$$\log(e^X e^Y) = X + \int_0^1 g(e^{\text{ad}_X} e^{t \text{ad}_Y})(Y) dt$$

NOTE : Observe that $e^{\text{ad}_X} e^{t \text{ad}_Y}$ and hence, also $g(e^{\text{ad}_X} e^{t \text{ad}_Y})$ are linear operators on the space $M_n(\mathbb{C})$ of all $n \times n$ complex matrices which is applied on the matrix Y . Here X and Y are assumed small implies that $e^{\text{ad}_X} e^{t \text{ad}_Y}$ is close to the identity operator on $M_n(\mathbb{C})$ for $0 \leq t \leq 1$, so that $g(e^{\text{ad}_X} e^{t \text{ad}_Y})$ is well defined.

Proof. For sufficiently small X and Y in $M_n(\mathbb{C})$, let

$$Z(t) = \log(e^X e^{tY})$$

for $0 \leq t \leq 1$. We will compute $Z(1)$. Since $e^{Z(t)} = e^X e^{tY}$, we have

$$e^{-Z(t)} \frac{d}{dt} e^{Z(t)} = (e^X e^{tY})^{-1} e^X e^{tY} Y = Y$$

On the other hand, by Theorem (1.4),

$$e^{-Z(t)} \frac{d}{dt} e^{Z(t)} = \left\{ \frac{I - e^{-\text{ad}_{Z(t)}}}{\text{ad}_{Z(t)}} \right\} \left(\frac{dZ}{dt} \right)$$

Hence,

$$\left\{ \frac{I - e^{-\text{ad}_{Z(t)}}}{\text{ad}_{Z(t)}} \right\} \left(\frac{dZ}{dt} \right) = Y$$

Now, if X and Y are small enough, $Z(t)$ will also be small, so that $T = [I - e^{-\text{ad}_{Z(t)}}] / \text{ad}_{Z(t)}$ will be close to the identity and so its eigenvalues close to 1 as $|\lambda - 1| < \|T - I\|$ and hence invertible. So, we get

$$\frac{dZ}{dt} = \left\{ \frac{I - e^{-\text{ad}_{Z(t)}}}{\text{ad}_{Z(t)}} \right\}^{-1} (Y)$$

Now, if X and Y are small enough, $Z(t)$ will also be small, so that $T = [I - e^{-\text{ad}_{Z(t)}}] / \text{ad}_{Z(t)}$ will be close to the identity and so its eigenvalues close to 1 as $|\lambda - 1| < \|T - I\|$ and hence invertible. So, we get

$$\frac{dZ}{dt} = \left\{ \frac{I - e^{-\text{ad}_{Z(t)}}}{\text{ad}_{Z(t)}} \right\}^{-1} (Y)$$

Also applying the Ad homomorphism formula to the equation $e^{Z(t)} = e^X e^{tY}$, we get :

$$\begin{aligned}\text{Ad}_{e^{Z(t)}} &= \text{Ad}_{e^X} \text{Ad}_{e^{tY}} \\ \implies e^{\text{ad}_{Z(t)}} &= e^{\text{ad}_X} e^{t \text{ad}_Y} \\ \implies \text{ad}_{Z(t)} &= \log(e^{\text{ad}_X} e^{t \text{ad}_Y}).\end{aligned}$$

Putting the last relations give us:

$$\frac{dZ}{dt} = \left\{ \frac{I - (e^{\text{ad}_X} e^{t \text{ad}_Y})^{-1}}{\log(e^{\text{ad}_X} e^{t \text{ad}_Y})} \right\}^{-1} (Y)$$

Now, notice that

$$g(z) = \left\{ \frac{1 - z^{-1}}{\log z} \right\}^{-1}$$

So that we get,

$$\frac{dZ}{dt} = g(e^{\text{ad}_X} e^{t \text{ad}_Y}) (Y).$$

Observing that $Z(0) = X$ and integrating gives us

$$\log(e^X e^Y) = Z(1) = X + \int_0^1 g(e^{\text{ad}_X} e^{t \text{ad}_Y}) (Y) dt$$

which is the Baker-Campbell-Hausdorff formula. $\square$

Corollary 2.2. (Series form of the BCH Formula) : For all $n \times n$ complex matrices X and Y with $\|X\|$ and $\|Y\|$ sufficiently small, we have

$$\log(e^X e^Y) = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] - \frac{1}{12}[Y, [X, Y]] + \dots$$

Proof. We use the analytic series of $g(z)$

$$g(z) = 1 + \frac{1}{2}(z - 1) - \frac{1}{6}(z - 1)^2 + \frac{1}{12}(z - 1)^3 - \dots$$

First of all,

$$\begin{aligned}e^{\text{ad}_X} e^{t \text{ad}_Y} - I &= \left(I + \text{ad}_X + \frac{(\text{ad}_X)^2}{2} + \dots \right) \left(I + t \text{ad}_Y + \frac{t^2 (\text{ad}_Y)^2}{2} + \dots \right) - I \\ &= \text{ad}_X + t \text{ad}_Y + t \text{ad}_X \text{ad}_Y + \frac{(\text{ad}_X)^2}{2} + \frac{t^2 (\text{ad}_Y)^2}{2} + \dots\end{aligned}$$

Since $e^{\text{ad}_X} e^{t \text{ad}_Y} - I$ has no zeroth-order term, $(e^{\text{ad}_X} e^{t \text{ad}_Y} - I)^m$ will contribute only terms of degree m or higher in ad_X or ad_Y . Computing up to degree 2 in ad_X and ad_Y gives

$$g(e^{\text{ad}_X} e^{t \text{ad}_Y}) = I + \frac{1}{2} \left(\text{ad}_X + t \text{ad}_Y + t \text{ad}_X \text{ad}_Y + \frac{(\text{ad}_X)^2}{2} + \frac{t^2 (\text{ad}_Y)^2}{2} \right)$$

$$-\frac{1}{6}[(\text{ad}_X)^2 + t^2(\text{ad}_Y)^2 + t\text{ad}_X\text{ad}_Y + t\text{ad}_Y\text{ad}_X] + \text{higher-order terms}.$$

Now apply $g(e^{\text{ad}_X}e^{t\text{ad}_Y})$ to Y and integrate. Computing to second order and noting that any term with ad_Y acting first with Y is zero, we get:

$$\begin{aligned} \log(e^X e^Y) &= X + \int_0^1 \left[Y + \frac{1}{2}[X, Y] + \frac{1}{4}[X, [X, Y]] - \frac{1}{6}[X, [X, Y]] - \frac{t}{6}[Y, [X, Y]] \right] dt + \dots \\ &= X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] - \frac{1}{12}[Y, [X, Y]] + \dots \end{aligned}$$

for first few terms. □

3 Connected Lie Subgroups

Definition. If H is any subgroup of $GL(n; \mathbb{C})$. Then,

$$\mathfrak{h} = \{X \in \mathfrak{g} : e^{tX} \in H, \quad \forall t \in \mathbb{R}\}$$

is **lie algebra** of H .

Definition. If G is a matrix Lie group with Lie algebra $\mathfrak{g}$, then $H \subset G$ is a **connected Lie subgroup** of G if the following conditions are satisfied:

1. H is a subgroup of G .
2. The Lie algebra $\mathfrak{h}$ of H is a Lie subalgebra of $\mathfrak{g}$.
3. Every element of H are in the form $e^{X_1}e^{X_2}\dots e^{X_m}$, with $X_1, \dots, X_m \in \mathfrak{h}$.

Observe that any group H as in the definition is path connected, since each element of H can be connected to the identity I in H by a path

$$t \mapsto e^{(1-t)X_1}e^{(1-t)X_2}\dots e^{(1-t)X_m}$$

Lemma 3.1. Choose a basis for $\mathfrak{h}$ and call an element of $\mathfrak{h}$ **rational** if its coefficients with respect to this basis are rational. Then for every $\delta > 0$ and every $A \in H$, there exist rational elements $R_1, \dots, R_m$ in $\mathfrak{h}$ and X in $\mathfrak{h}$ with $\|X\| < \delta$, such that

$$A = e^{R_1}e^{R_2}\dots e^{R_m}e^X.$$

Proof. Choose $\varepsilon > 0$ such that $\forall X, Y \in \mathfrak{h}$ with $\|X\| < \varepsilon$ and $\|Y\| < \varepsilon$, the Baker-Campbell-Hausdorff formula holds for X and Y . Let $C(\cdot, \cdot)$ denote the right-hand side of the formula, so that

$$e^X e^Y = e^{C(X, Y)}$$

whenever $\|X\|, \|Y\| < \varepsilon$. Also $C(\cdot, \cdot)$ is continuous in X, Y . So for a given ε there exists some $\delta > 0$ such that for $\|X\|, \|Y\| < \delta$, implies $\|C(X, Y)\| < \varepsilon$. So, we can assume $\delta < \varepsilon$. Since $e^X = (e^{X/k})^k$, we can write every element A of H in the form

$$A = e^{X_1}\dots e^{X_N}$$

with $X_j \in \mathfrak{h}$ and $\|X_j\| < \delta$.

We now apply induction on N . If $N = 0$, then $A = I = e^0$, so nothing to prove. Assume the lemma for A 's that can be expressed in the above form for some integer N , and consider A of the form

$$A = e^{X_1} \dots e^{X_N} e^{X_{N+1}}$$

with $X_j \in \mathfrak{h}$ and $\|X_j\| < \delta$. Applying our induction hypothesis to $e^{X_1} \dots e^{X_N}$, we obtain

$$\begin{aligned} A &= e^{R_1} \dots e^{R_m} e^X e^{X_{N+1}} \\ &= e^{R_1} \dots e^{R_m} e^{C(X, X_{N+1})} \end{aligned}$$

where the R_j 's are rational and $\|C(X, X_{N+1})\| < \varepsilon$. Since $\mathfrak{h}$ is a subalgebra of $\mathfrak{gl}(n; \mathbb{C})$, $C(X, X_{N+1})$ is in $\mathfrak{h}$, but it may not have norm less than δ . So choose a rational element R_{m+1} of $\mathfrak{h}$ that is close to $C(X, X_{N+1})$ with $\|R_{m+1}\| < \varepsilon$ as rational points are dense in $\mathfrak{h}$. So we get,

$$\begin{aligned} A &= e^{R_1} \dots e^{R_m} e^{R_{m+1}} e^{-R_{m+1}} e^{C(X, X_{N+1})} \\ &= e^{R_1} \dots e^{R_m} e^{R_{m+1}} e^{X'} \end{aligned}$$

where $X' = C(-R_{m+1}, C(X, X_{N+1}))$

Then X' will be in $\mathfrak{h}$, and by choosing R_{m+1} sufficiently close to $C(X, X_{N+1})$, we can make $\|X'\| < \delta$. Since $C(-Z, Z) = \log(e^{-Z}e^Z) = 0$ for all small Z , and if Z' is close to Z , then $C(-Z', Z)$ will be more smaller. □

Discussion 3.2.

Think of $\mathfrak{gl}(n; \mathbb{C})$ as $\mathbb{R}^{2n^2}$ and decompose $\mathfrak{gl}(n; \mathbb{C})$ as $\mathfrak{h} \oplus D$, where D is the orthogonal complement of $\mathfrak{h}$ with respect to the usual inner product on $\mathbb{R}^{2n^2}$ as show in in the proof of Theorem (1.3). Then by the Inverse Function Theorem, there exist neighborhoods U and V of the origin in $\mathfrak{h}$ and D , respectively, and a neighborhood W of I in $\text{GL}(n; \mathbb{C})$ with the property that for each $A \in W$ we can write it uniquely as

$$A = e^X e^Y, \quad X \in U, Y \in V \quad (3.2)$$

in such a way that X and Y depend continuously on A .

Lemma 3.3. *Following the above discussion let V be a neighborhood of the origin in D as in (3.2). If $E \subset V$ is defined as*

$$E = \{Y \in V \mid e^Y \in H\}$$

Then E is at most countable.

Proof. Fix δ such that for all X, Y with $\|X\|, \|Y\| < \delta$, the quantity $C(X, Y)$ is defined and contained in U . Our claim is that for each sequence $R_1, \dots, R_m$ of rational elements in $\mathfrak{h}$, there is at most one $X \in \mathfrak{h}$ with $\|X\| < \delta$ such that the element

$$e^{R_1} e^{R_2} \dots e^{R_m} e^X \quad (3.3)$$

belongs to e^V . Since, if we have

$$\begin{aligned} e^{R_1} e^{R_2} \dots e^{R_m} e^{X_1} &= e^{Y_1}, \\ e^{R_1} e^{R_2} \dots e^{R_m} e^{X_2} &= e^{Y_2} \end{aligned}$$

with $Y_1, Y_2 \in V$, then

$$e^{-X_1} e^{X_2} = e^{-Y_1} e^{Y_2}$$

and so

$$e^{-Y_1} = e^{-X_1} e^{X_2} e^{-Y_2} = e^{C(-X_1, X_2)} e^{-Y_2}$$

with $C(-X_1, X_2) \in U$. However, each element of $e^U e^V$ has a unique representation as $e^Y e^X$ with $X \in U$ and $Y \in V$. Thus, we must have $-Y_2 = -Y_1$ and so, $e^{X_1} = e^{X_2}$ and $X_1 = X_2$. By Lemma (3.1), every element of H can be expressed in the (3.1) lemma form with $\|X\| < \delta$. Now, there are only countably many rational elements in $\mathfrak{h}$ and thus only countably many expressions of the form $e^{R_1} \dots e^{R_m}$, each of which produces at most one element of the (3.3) form that belongs to e^V . Thus, the set E in Lemma (3.3) is at most countable. $\square$

Theorem 3.4. *Let G be a matrix Lie group with Lie algebra $\mathfrak{g}$ and let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$. Then there exists a unique connected Lie subgroup H of G with Lie algebra $\mathfrak{h}$.*

Proof. It is enough to assume that $G = \text{GL}(n; \mathbb{C})$. Indeed, if G is a matrix lie group of $\text{GL}(n; \mathbb{C})$ and $\mathfrak{h}$ is a Lie subalgebra of $\mathfrak{g}$ then applying the theorem for $\text{GL}(n; \mathbb{C})$ there exists H a connected Lie subgroup of $\text{GL}(n; \mathbb{C})$ as $\mathfrak{h}$ is a lie subalgebra in $\mathfrak{gl}(n; \mathbb{C})$. And for every A in H written as $A = e^{X_1} e^{X_2} \dots e^{X_m}$ with $X_i \in \mathfrak{h} \subset \mathfrak{g}$ we get that $A \in G$. Hence H is a connected lie subgroup of G . Now let

$$H = \{e^{X_1} e^{X_2} \dots e^{X_N} \mid X_1, \dots, X_N \in \mathfrak{h}\}$$

which is a subgroup of G . All we need to prove that the Lie algebra of H is $\mathfrak{h}$.

Let $\mathfrak{h}'$ be the Lie algebra of H . Then $\mathfrak{h} \subseteq \mathfrak{h}'$. For $Z \in \mathfrak{h}'$, we may write, for all sufficiently small t ,

$$e^{tZ} = e^{X(t)} e^{Y(t)} \quad (3.4)$$

where $X(t) \in U \subset \mathfrak{h}$ and $Y(t) \in V \subset D$ and $e^{tZ} \in W \subset H$ by using (3.2). Note that $X(t)$ and $Y(t)$ are continuous functions of t as by the local inverse function (which exists due to the inverse function theorem), $X(t), Y(t)$ depends on tZ continuously and hence on t since Z is fixed. Since $Z \in \mathfrak{h}$, we have $e^{tZ} \in H$ for all t . Also $e^{X(t)} \in H$, hence $e^{Y(t)} \in H$ for all sufficiently small t . If $Y(t)$ is not constant, then it would take on uncountably many values, implying E is uncountable, contradicting Lemma (3.3). So, $Y(t)$ must be constant. Since

$Y(0) = 0$ which comes from the uniqueness of the above equation (3.4), $Y(t) \equiv 0$. Thus, for small t , we get $e^{tZ} = e^{X(t)}$ implying $tZ = X(t) \in \mathfrak{h}$. So, $Z \in \mathfrak{h}$ implying $\mathfrak{h}' \subseteq \mathfrak{h}$. □

4 Ado's theorem

Definition. If L is a Lie algebra over the field F , the universal enveloping algebra $\mathcal{U}$ of L is an associative F -algebra with 1 together with a linear map $i : L \rightarrow \mathcal{U}$ satisfying $i([x, y]) = i(x)i(y) - i(y)i(x)$ for $x, y \in L$, and has the following universal property: If $\mathfrak{A}$ is any F -algebra with 1, and $j : L \rightarrow \mathfrak{A}$ is a linear map satisfying $j([x, y]) = j(x)j(y) - j(y)j(x)$, then there exists unique F -algebra homomorphism $\psi : \mathcal{U} \rightarrow \mathfrak{A}$ such that $j = \psi \circ i$. In other words, the following diagram commutes:

$$\begin{array}{ccc}
 L & & \\
 \downarrow i & \searrow j & \\
 \mathcal{U} & \xrightarrow{\psi} & \mathfrak{A}
 \end{array}$$

Theorem 4.1. *The canonical map $i : L \rightarrow \mathcal{U}(L)$ is injective.*

Theorem 4.2. Poincare-Birkhoff-Witt Theorem : *Let $(x_1, x_2, \dots)$, be any ordered basis of L . Let $y_n = i(x_n) \in \mathcal{U}$. Then the elements $y_{j(1)} \dots y_{j(m)}$, $m \in \mathbb{N}$ and $j(1) \leq j(2) \leq \dots \leq j(m)$ together with 1 form a basis of $\mathcal{U}$.*

Lemma 4.3. *If δ is a derivation of L , then there exists a derivation $\tilde{\delta}$ of $\mathcal{U}$ such that $\tilde{\delta} \circ i = i \circ \delta$*

Lemma 4.4. *Suppose S is a finite-dimensional solvable Lie algebra over F where $\text{char}(F) = 0$. Let $\mathcal{U}$ be the universal enveloping algebra of S , together with its linear map $i : S \rightarrow \mathcal{U}$. Suppose I is an ideal of $\mathcal{U}$ satisfying the following conditions: (i) $\mathcal{U}/I$ has a finite dimension. (ii) $i(\overline{x})$ is nilpotent in $\mathcal{U}/I$ for every $x \in \text{Nil}(\mathfrak{S})$.*

Then there exists an ideal J of $\mathcal{U}$ with $J \subseteq I$ such that J satisfies (1) and (2), and also,

(iii) If δ is a derivation of S , the corresponding derivation $\tilde{\delta}$ of $\mathcal{U}$ satisfies $D(J) \subseteq J$.

Definition. We call a representation $\rho : L \rightarrow \mathfrak{gl}(V)$ a nilrepresentation if $\rho(x)$ is a nilpotent endomorphism for each $x \in \text{Nil}(L)$.

Theorem 4.5. Levi Decomposition Theorem: Let $\mathfrak{g}$ be a lie algebra over a field F and S be its radical. Then $\mathfrak{g}$ admits a Levi subalgebra that means, there exists a lie subalgebra H in $\mathfrak{g}$ s.t

$$\mathfrak{g} = S \oplus H$$

Note that H is a semisimple lie algebra

Proposition 1. Let $L = S \oplus H$ be a finite-dimensional Lie algebra where S is a solvable ideal and H is a subalgebra of L . Suppose $\sigma : S \rightarrow \mathfrak{gl}(V)$ be a finite-dimensional nilrepresentation. Then there exists a finite-dimensional representation $\rho : L \rightarrow \mathfrak{gl}(V')$ such that $\ker(\rho) \cap S \subseteq \ker(\sigma)$. Also, if $\text{Nil}(L) = \text{Nil}(S)$ or $\text{Nil}(L) = L$, then ρ can be chosen to be a nilrepresentation.

Theorem 4.6. Ado's Theorem: Let L be a finite-dimensional Lie algebra over a field of characteristic 0. Then L has a faithful finite-dimensional representation.

Proof. Let $Z(L)$ denote the centre of L . Let $z_1, z_2, \dots, z_n$ be a basis for $Z(L)$. We claim that the map $\rho_0 : Z(L) \rightarrow \mathfrak{gl}(F^{2n})$ defined by

$$\rho_0(c_1 z_1 + c_2 z_2 + \dots + c_n z_n) : (a_1, b_1, a_2, b_2, \dots, a_n, b_n) \mapsto (c_1 b_1, 0, c_2 b_2, 0, \dots, c_n b_n, 0)$$

is a faithful finite-dimensional nilrepresentation of $Z(L)$. Indeed if

$$\rho_0(c_1 z_1 + c_2 z_2 + \dots + c_n z_n) = \rho_0(d_1 z_1 + d_2 z_2 + \dots + d_n z_n)$$

, then

$$(c_1 b_1, 0, c_2 b_2, 0, \dots, c_n b_n, 0) = (d_1 b_1, 0, d_2 b_2, 0, \dots, d_n b_n, 0)$$

for every $b_1, b_2, \dots, b_p \in F$. Setting $b_i = 1$ and $b_j = 0$ for all $i \neq j$, we get $c_i = d_i$. So ρ_0 is faithful.

To check that ρ_0 is a representation, we need to prove $\rho_0([x, y]) = [\rho_0(x), \rho_0(y)]$ for every $x, y \in Z(L)$. Since $Z(L)$ is commutative, this is equivalent to showing $[\rho_0(x), \rho_0(y)] = 0$, i.e. $\rho_0(x) \circ \rho_0(y) = \rho_0(y) \circ \rho_0(x)$. Let $x = c_1 z_1 + \dots + c_n z_n$ and $y = d_1 z_1 + \dots + d_n z_n$. Then,

$$\begin{aligned} \rho_0(x) \circ \rho_0(y) : (a_1, b_1, a_2, b_2, \dots, a_n, b_n) &\xrightarrow{\rho_0(y)} (d_1 b_1, 0, d_2 b_2, 0, \dots, d_n b_n, 0) \\ &\xrightarrow{\rho_0(x)} (0, 0, 0, 0, \dots, 0, 0) \end{aligned}$$

and similarly,

$$\begin{aligned} \rho_0(y) \circ \rho_0(x) : (a_1, b_1, a_2, b_2, \dots, a_n, b_n) &\xrightarrow{\rho_0(x)} (c_1 b_1, 0, c_2 b_2, 0, \dots, c_n b_n, 0) \\ &\xrightarrow{\rho_0(y)} (0, 0, 0, 0, \dots, 0, 0) \end{aligned}$$

So $\rho_0(x) \circ \rho_0(y) = \rho_0(y) \circ \rho_0(x) = 0$ as required. In particular, $(\rho_0(x))^2 = 0$ for all $x \in Z(L)$, which shows that ρ_0 is a nilrepresentation. Because $\text{Rad}(L)$ is solvable, and $Z(L) \subseteq \text{Rad}(L)$ there exists a sequence of solvable subalgebras (in fact ideals) in L

$$Z(L) = L_0 \subseteq L_1 \subseteq \dots \subseteq L_k = \text{Nil}(L) \subseteq L_{k+1} \subseteq \dots \subseteq L_m = \text{Rad}(L) \subseteq L_{m+1} = L$$

such that L_i is an ideal in L_{i+1} for $0 \leq i \leq m$, and $\dim(L_{i+1}) = \dim(L_i) + 1$ for $0 \leq i \leq m-1$. Hence, $L_{i+1} = L_i \oplus Fv_i$ where $v_i \in L_{i+1} \setminus L_i$ for each $0 \leq i \leq m-1$. And for the case $i = m$, we use Levi's Theorem to get $L = \text{Rad}(L) \oplus H$ for some semisimple subalgebra H of L .

Starting with the nilrepresentation ρ_0 of $Z(L) = L_0$, we apply the last Proposition inductively to get a representation ρ_1 of L_1 such that $\ker(\rho_1) \cap L_0 \subseteq \ker(\rho_0)$. For the induction to work, we need each ρ_i is nilrepresentation which is true because $\text{Nil}(L_i) = L_i$ for $1 \leq i \leq k$ (as $L_i \subseteq \text{Nil}(L)$ for $0 \leq i \leq k$) and $\text{Nil}(L_i) = \text{Nil}(L)$ for $i \geq k$ (as $\text{Nil}(L)$ is the maximal nilpotent ideal of L). Note that from $Z(L) = Z(L) \cap L_i$ for all $0 \leq i \leq m+1$ we get

$$Z(L) \cap \ker(\rho_i) = Z(L) \cap L_{i-1} \cap \ker(\rho_i) \subseteq Z(L) \cap \ker(\rho_{i-1})$$

for all $1 \leq i \leq m+1$. Since $Z(L) \cap \ker(\rho_0) = 0$ (because ρ is faithful), it follows that $Z(L) \cap \ker(\rho_i) = 0$ for all $1 \leq i \leq m+1$. Hence, $Z(L) \cap \ker(\rho_{m+1}) = 0$. We have the representation $\rho_{m+1} : L_{m+1} = L \rightarrow \mathfrak{gl}(V)$ for some vector space with $\dim(V) < \infty$. Define $\rho = \rho_{m+1} \oplus \text{ad} : L \rightarrow \mathfrak{gl}(V \oplus L)$ by $\rho(x) = (\rho_{m+1}(x), \text{ad}(x))$. Then ρ is a finite-dimensional representation of L satisfying

$$\ker(\rho) = \ker(\rho_{m+1}) \cap \ker(\text{ad}) = \ker(\rho_{m+1}) \cap Z(L) = 0$$

so ρ is faithful. This concludes the proof of Ado's Theorem □

5 Lie's Third Theorem

Theorem 5.1. *If $\mathfrak{g}$ is any finite dimensional real Lie Algebra, there exists a connected Lie subgroup G of $GL(n; \mathbb{C})$ whose lie algebra is isomorphic to $\mathfrak{g}$*

Proof. By Ado's theorem, we identify $\mathfrak{g}$ with a real lie subalgebra of $\mathfrak{gl}(n; \mathbb{C})$. Hence, by theorem (3.4), there is a connected lie subgroup of $GL(n; \mathbb{C})$ with lie algebra $\mathfrak{g}$. □

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